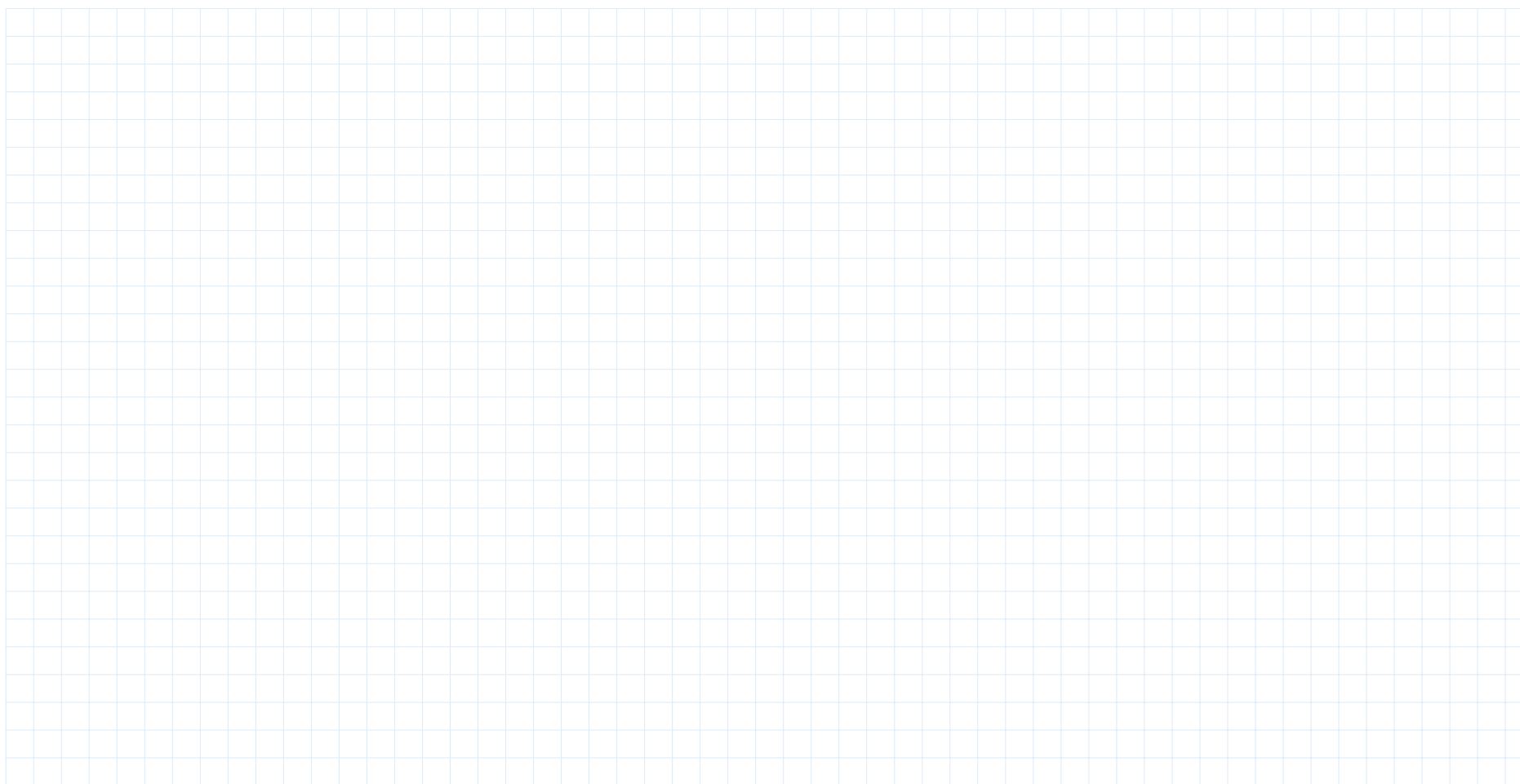


The value $\arcsin x = \sin^{-1} x$ is the angle in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ whose sine is x . Its input must satisfy $-1 \leq x \leq 1$. The exponent -1 here means inverse, not reciprocal. **Example.** Find the exact value and explain your choice of angle:

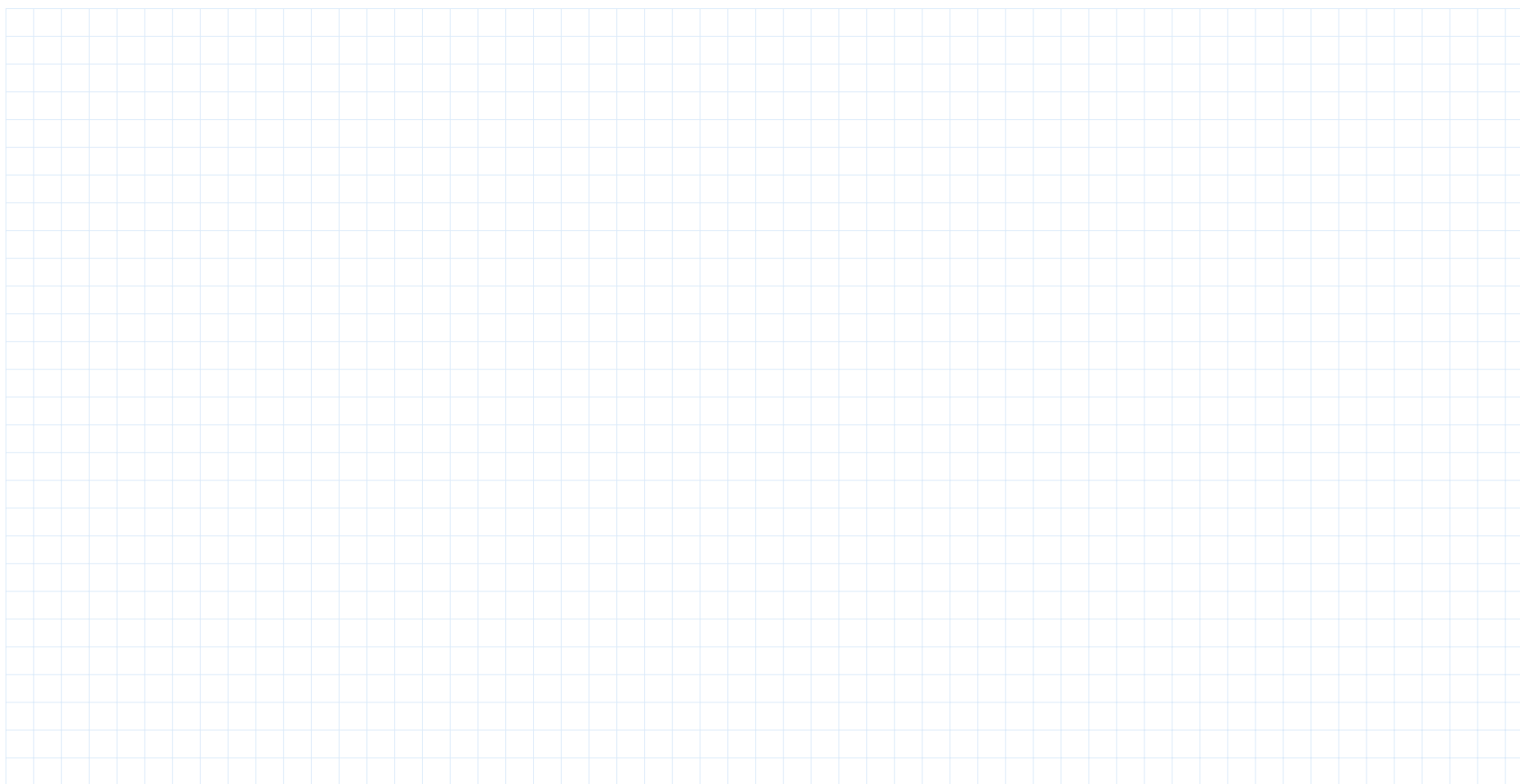
$$\arcsin\left(-\frac{1}{2}\right).$$



Inverse Cosine and Tangent

$\arccos x$ returns an angle in $[0, \pi]$ and has domain $[-1, 1]$. $\arctan x$ returns an angle in $(-\frac{\pi}{2}, \frac{\pi}{2})$ and has domain $\mathbb{R}$. Find each exact value. Use the required range to choose the angle.

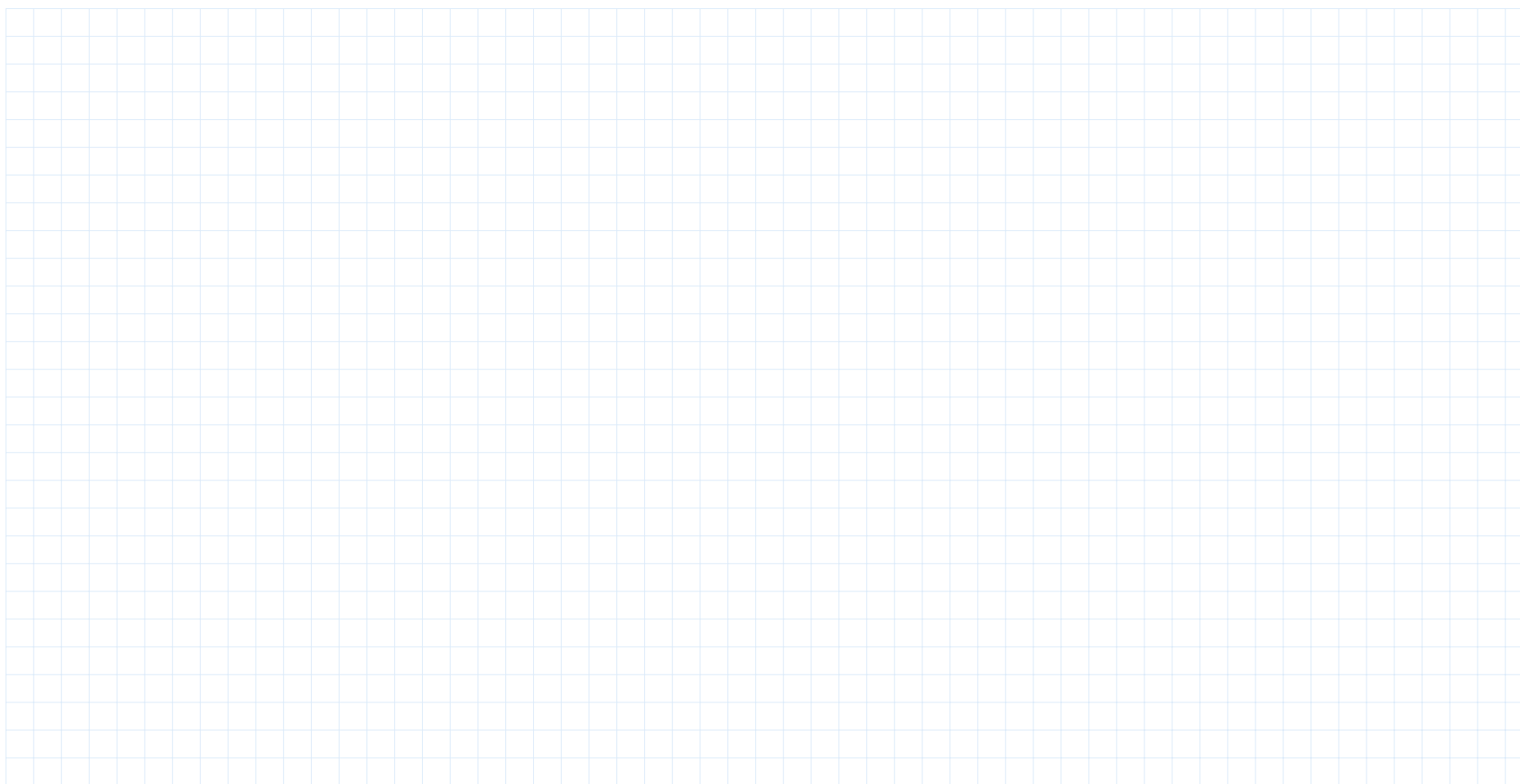
(a) $\arccos\left(-\frac{\sqrt{2}}{2}\right)$ (b) $\arctan(-\sqrt{3})$.



Compositions: Check the Range

An inverse reverses its original function only on the selected domain. Find each exact value; do not cancel the function symbols without checking.

(a) $\sin\left(\arcsin\left(-\frac{3}{5}\right)\right)$ (b) $\arcsin\left(\sin\frac{5\pi}{6}\right).$

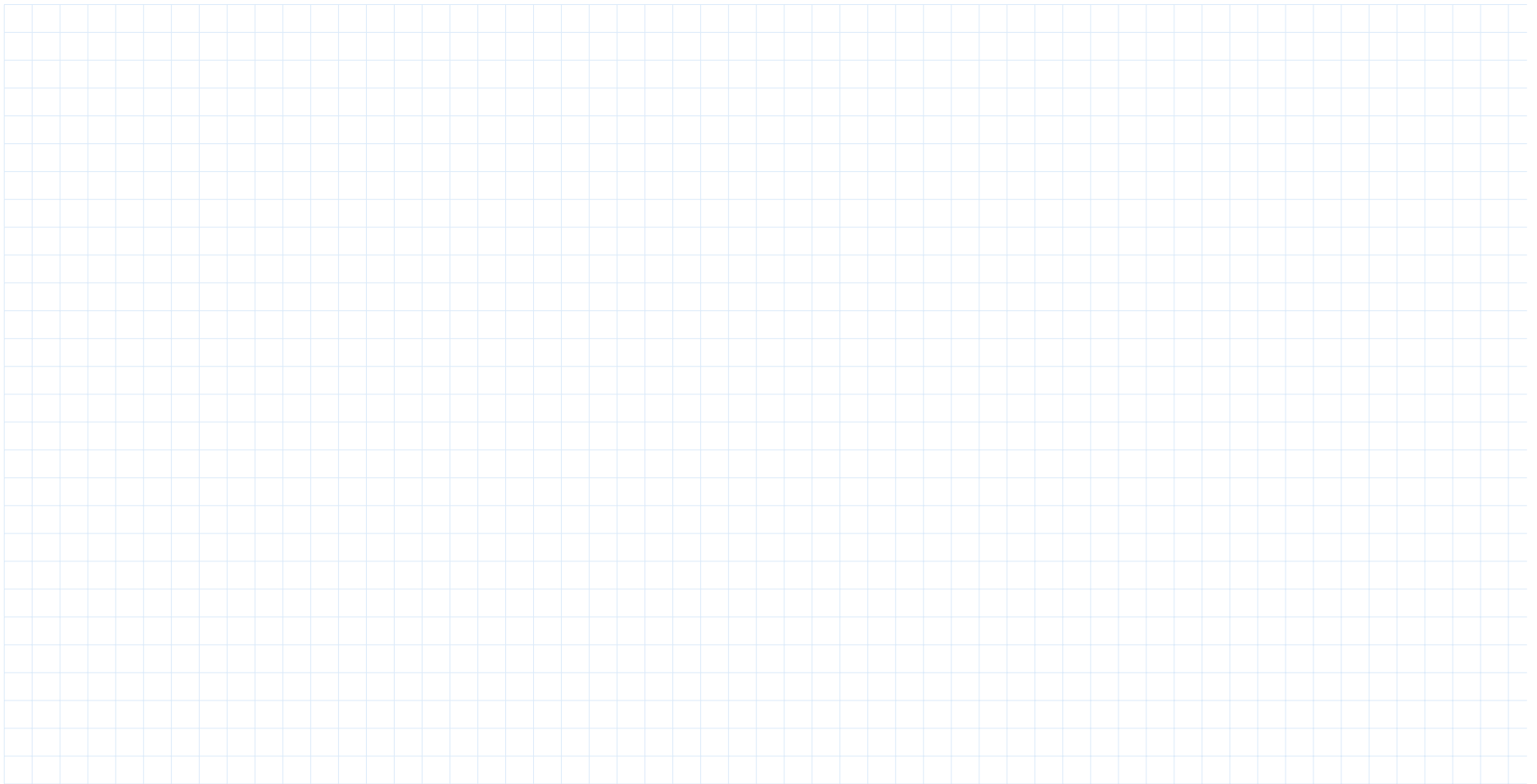


Approximate Inverse Values

Use a calculator in **radian mode**. Round to two decimal places.

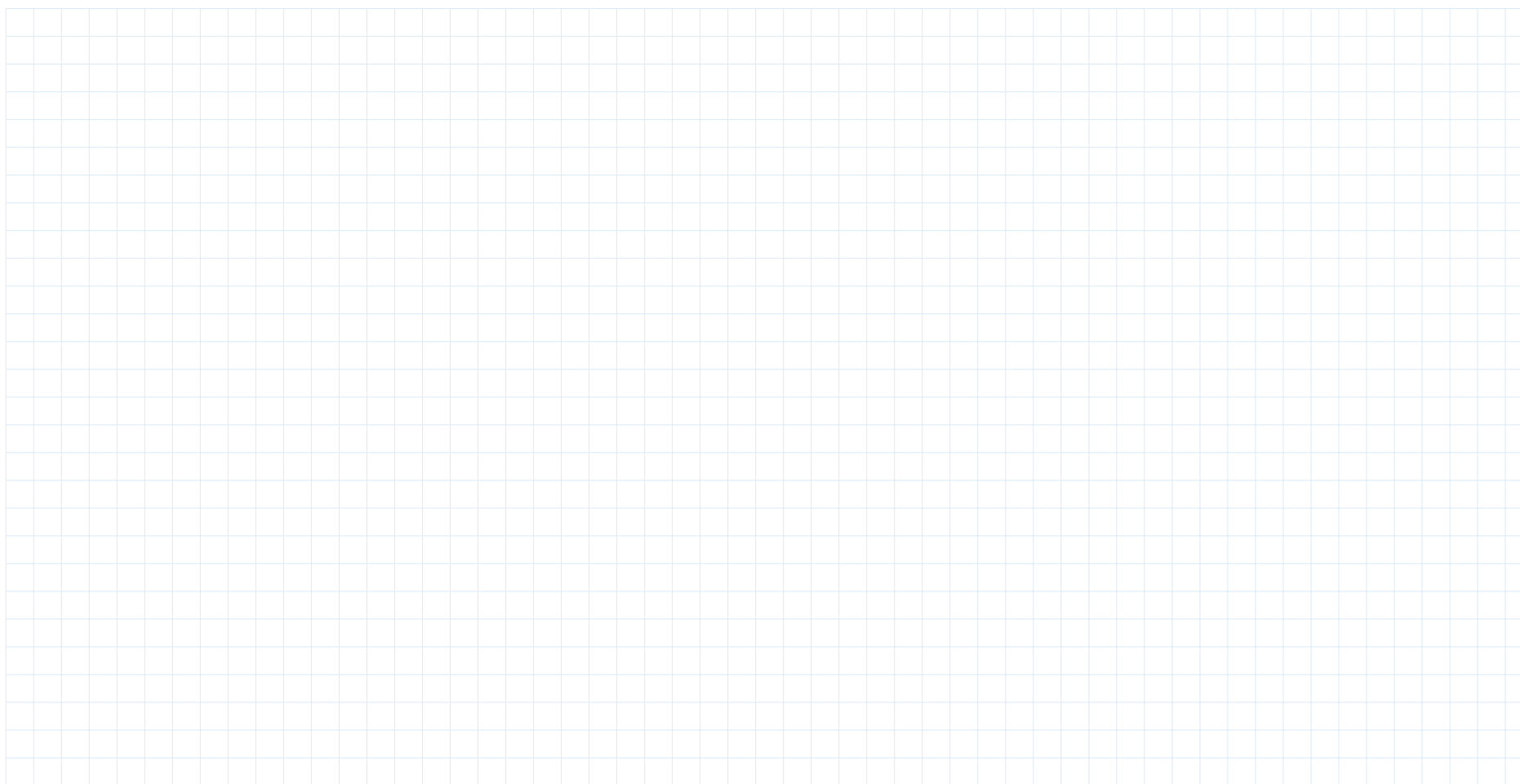
(a) $\arcsin(0.7)$ (b) $\arctan(-2)$.

For each answer, check whether its sign and size fit the principal range.



Find an Inverse Function

Let $f(x) = 2 \sin x - 1$ with the restricted domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Find $f^{-1}(x)$. State the domain and range of the inverse.



Equations with Inverse Functions

Solve each equation over the real numbers. Check the allowable output of the inverse function.

(a) $4 \arccos x = 3\pi$

(b) $6 \arcsin x = 4\pi$.

